In Exercises 29–32, find the elementary row operation that transforms the first matrix into the second, and then find the reverse row ~~rotation~~ operation that transforms into the first matrix from the second matrix.

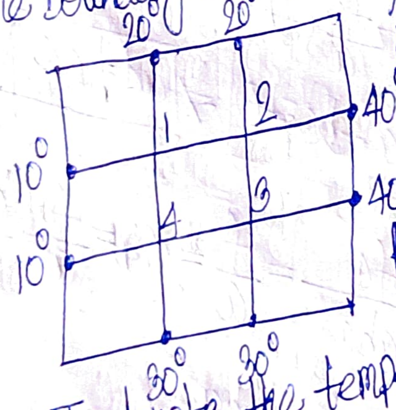
29) $\begin{bmatrix} 0 & -2 & 5 \\ 1 & 4 & -7 \\ 3 & -1 & 6 \end{bmatrix}$, $\begin{bmatrix} 1 & 4 & -7 \\ 0 & -2 & 5 \\ 3 & -1 & 6 \end{bmatrix}$

First to Second: Interchange rows 1 and 2
Second to First: Interchange rows 1 and 2

31) $\begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 5 & -2 & 8 \\ 4 & -1 & 3 & -6 \end{bmatrix}$, $\begin{bmatrix} 1 & -2 & 1 & 0 \\ 0 & 5 & -2 & 8 \\ 0 & 7 & -1 & 6 \end{bmatrix}$

First to Second: Row 3 = Row 3 − 4 × Row 1
Second to First: Row 3 = Row 3 + 4 × Row 1

An important concern in the study of heat transfer is to determine the steady-state distribution of a thin plate when the temperature around the boundary is known. Assume the plate shown in the figure represents a cross section of a metal beam, with negligible heat flow in the direction perpendicular to the plate.



Let $T_1, \ldots, T_4$ denote the temperatures at the four interior nodes of the mesh in the figure. The temperature at a node is approximately equal to the average of the four nearest nodes — to the left, above, to the right, and below. For instance,

$T_1 = (10 + 20 + T_2 + T_4)/4$, or $4T_1 - T_2 - T_4 = 30$